

1. Noah placed an empty bowl on a scale. He added different amounts of a liquid to the bowl, recording the amount of liquid added, in tablespoons, and the total weight of the bowl and liquid, in grams, each time on a scatter plot. He fit the line $y = 14.75x + 861.82$ to the data in his scatter plot, where x represents the amount of the liquid, in tablespoons, and y represents the total weight of the bowl and the liquid, in grams.

a. Interpret the slope of the line based on the context.

b. Interpret the y -intercept of the line based on the context.

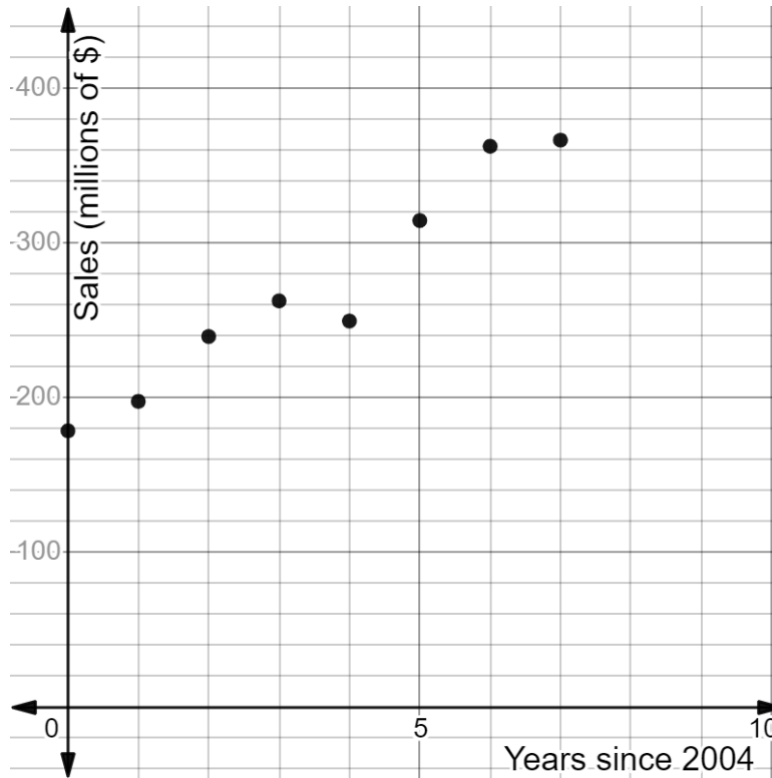
- c. Determine the weight, in grams, when 18 tablespoons of the liquid are placed in the bowl.

Weight (grams)	
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- d. Determine the approximate number of tablespoons of the liquid Noah needs to add so that the total weight is 1,200 grams.

Amount of Liquid (tablespoons)	
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2. The total sales, in millions of dollars, of Florida strawberries for the years 2004 – 2011 is shown on the coordinate grid.



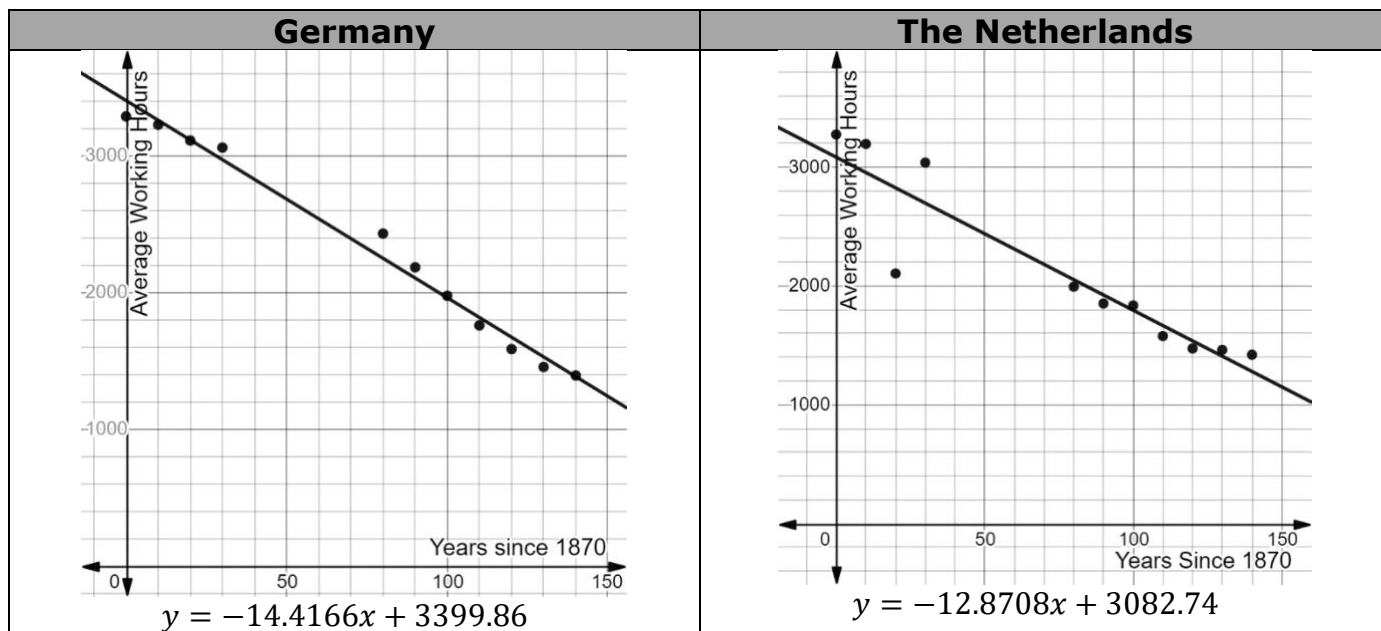
- a. Draw a trend line on the scatter plot.
- b. Use your trend line to write an equation that would fit the data.

Equation	
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- c. Interpret the slope of the line based on the context.
- d. Interpret the y -intercept of the line based on the context.

3. A social scientist researched how the average working hours per worker over a full year for two different European countries changed since 1870. She created a scatter plot and found the equation for the line of best fit for each set of data. She then used this equation to create a table of values for the residuals.

The scatter plots, equations for the lines of best fit, and residuals for the two countries are shown, where x represents the number of years since 1870 and y represents the average working hours per worker over a full year.



Years since 1870	Residuals
0	-115.86
10	-32.694
20	-3.528
30	88.638
80	180.8967
90	78.7563
100	14.3177
110	-57.9257
120	-87.2645
130	-73.6891
140	8.3469

Years since 1870	Residuals
0	191.26
10	239.968
20	279.676
30	340.384
80	-59.0811
90	-72.1563
100	39.9225
110	-88.5915
120	-64.3725
130	52.4852
140	140.5132

a. Complete the statement.

The data for The Netherlands has a

- ☐ positive
- ☐ negative

correlation because

- ☐ the slope of the line of best fit is
- ☐ most of the residual values are

- ☐ positive.
- ☐ negative.

b. Complete the statement.

☐ Germany

☐ The Netherlands

 has a

☐ weaker

☐ stronger

 correlation than

☐ Germany

☐ The Netherlands

because the residuals have a

☐ larger

☐ smaller

 absolute value, indicating that the

distance each

☐ actual

☐ predicted

 value is from the line of best fit is

☐ larger.

☐ smaller.

4. Each year, the State of Florida collects data on the price per box for each type of orange. A mathematician for a grower's association analyzed the data for the years 2009 to 2018 for a certain variety of orange. Some of the data is shown.

Years Since 2009	0	1	2	3	4	5	6	7
Price (\$)	5.95	7.11	8.88	6.25	8.41	8.40	8.99	10.50

The mathematician found the equation $y = 0.5x + 6.31$ models the data.

Use the model to determine the price, in dollars, in 2018.

Price in 2018	\$
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